

Improving coverage accuracy of nonparametric prediction intervals

Peter Hall and Andrew Rieck

Australian National University, Canberra, Australia

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Summary. Methods are suggested for improving the coverage accuracy of intervals for predicting future values of a random variable drawn from a sampled distribution. It is shown that properties of solutions to such problems may be quite unexpected. For example, the bootstrap and the jackknife perform very poorly when used to calibrate coverage, although the jackknife estimator of the true coverage is virtually unbiased. A version of the smoothed bootstrap can be employed for successful calibration, however. Interpolation among adjacent order statistics can also be an effective way of calibrating, although even there the results are unexpected. In particular, whereas the coverage error can be reduced from $O(n^{-1})$ to orders $O(n^{-2})$ and $O(n^{-3})$ (where n denotes the sample size) by interpolating among two and three order statistics respectively, the next two orders of reduction require interpolation among five and eight order statistics respectively.

Keywords: Bootstrap; Calibration; Density estimation; Interpolation; Jackknife; Monte Carlo methods; Order statistic; Quantile; Smoothed bootstrap

1. Introduction

A sample quantile can be regarded either as an estimator of the corresponding population quantile or as the end point of an interval for predicting the value of another, independent, variable drawn from the sampled population. We introduce new methods for quantile estimation, optimized from the viewpoint of improving the coverage accuracy of the prediction interval. The problem is shown to exhibit several highly non-standard features. For example, whereas appropriate interpolation among two or three nearby order statistics produces successive reductions in coverage error, from $O(n^{-1})$ to $O(n^{-2})$ and $O(n^{-3})$ respectively, where n denotes the sample size, passing to a fourth-order statistic does not follow the same pattern. It is necessary to interpolate among five order statistics to achieve $O(n^{-4})$ accuracy, and among eight order statistics to attain $O(n^{-5})$ accuracy. These techniques, based as they are on linear combinations of order statistics, have points in common with L -estimation; see Hosking (1998) for a survey of the latter topic.

Interpolation among as many as five order statistics is an unattractive method, since for small samples it is likely to rely too heavily on asymptotic properties. It would generally be preferable to calibrate lower order methods to enhance performance. This is a standard approach in the context of confidence intervals, with double-bootstrap methods being a popular technique for calibration. See for example Hall (1992), page 20 and following feature, Efron and Tibshirani (1993), page 263 and following feature, Shao and Tu (1995),

Address for correspondence: Peter Hall, Centre for Mathematics and Its Applications, Australian National University, Canberra, ACT 0200, Australia.
E-mail: Peter.Hall@maths.anu.edu.au

page 155 and following feature, and Davison and Hinkley (1997), page 223 and following feature. Bootstrap methods for prediction in nonparametric contexts have been studied by, for example, Stine (1985), Bai and Olshen (1988), Bai *et al.* (1990), Beran (1990, 1992) and Kabaila (1993). However, the standard bootstrap performs poorly when used to calibrate in the present setting, not least because the quantiles of a bootstrap resample are likely to include tied values.

In the context of the calibration of prediction intervals the jackknife holds far more promise than the bootstrap, since the jackknife estimator of coverage probability is unbiased for the true coverage in a sample of size $n - 1$. Indeed, arguments based on Taylor expansions show that, perhaps after appropriate interpolation, jackknife calibration should reduce the order of magnitude of the coverage error. However, this turns out to be dramatically incorrect; far from reducing the coverage error, jackknife calibration actually increases it, by an order of magnitude. The difficulty is that the jackknife estimator of the amount of calibration that is required is highly correlated with the quantity that is being corrected, and this dependence adds a new, low order term to the expansion of coverage error. (Details are given in a longer article on which this paper is based; see Hall and Rieck (2000).) This problem is not alleviated by applying the jackknife correction to a sample of size $n - 1$, for example by discarding one of the original data at random before applying the correction.

The failure of the jackknife calibration approach is also unusual because coverage error in nonparametric prediction is closely related to bias in nonparametric estimation. Indeed, the problems of reducing either of these quantities can be formulated in almost identical ways. The jackknife solves the bias reduction problem very effectively but fails spectacularly in the case of calibration of prediction intervals.

We show that a smoothed version of the bootstrap can, however, be used for effective calibration. A consideration of the smoothed bootstrap goes back to Silverman and Young (1987) and Young (1988). In many problems, even the optimal choice of the degree of smoothing provides only a small improvement in performance, typically in a term whose size is asymptotically negligible. In the present setting, however, an appropriate choice of the smoothing parameter reduces the coverage error by an order of magnitude. We suggest an empirical method for selecting the smoothing parameter.

The performance of our methods is described through their theoretical properties, outlined in Section 2, and comparative numerical studies, summarized in Section 3. Derivations of the $O(n^{-3})$ coverage accuracy of three-point interpolation, and of the fact that four-point interpolation fails to give $O(n^{-4})$ accuracy, are deferred to Appendix A. Proofs of related results are given by Hall and Rieck (2000).

2. Methodology

2.1. Interpolation methods

Let $\mathcal{X} = \{X_1, \dots, X_n\}$ denote a random sample from the distribution with distribution function F , and let $\xi_\alpha = F^{-1}(\alpha)$ denote the distribution's α th quantile, assumed uniquely defined with $\alpha \in (0, 1)$. One estimator of ξ_α is $\tilde{\xi}_\alpha$, equal to the $[n\alpha]$ th largest datum in $\mathcal{X}$, where $[x]$ denotes the smallest integer greater than or equal to x . It is well known, and follows from Edgeworth expansion of the distributions of order statistics (e.g. Reiss (1989)), that $(-\infty, \tilde{\xi}_\alpha)$ has a coverage error of order n^{-1} as a prediction interval for future data values:

$$P(X \leq \tilde{\xi}_\alpha) = \alpha + O(n^{-1}), \quad (2.1)$$

where X denotes a random variable independent of $\mathcal{X}$ but drawn from the same population.

The remainder $O(n^{-1})$ at equation (2.1) is least possible, in the sense that, if ξ_α is uniquely defined and the sampling distribution has a bounded density, not only does expression (2.1) hold but

$$\limsup_{n \rightarrow \infty} \{n|P(X \leq \tilde{\xi}_\alpha) - \alpha|\} > 0.$$

Nevertheless, if α is exactly equal to an integer divided by $n + 1$ then $P(X \leq \tilde{\xi}_\alpha) = \alpha$ exactly. More generally, linear interpolation of $\tilde{\xi}_\alpha$ reduces the coverage error to $O(n^{-2})$, i.e. if we choose the integer m so that $m \leq (n + 1)\alpha < m + 1$, and define

$$\tilde{\xi}_\alpha = \{m + 1 - (n + 1)\alpha\}X_{n,m} + \{(n + 1)\alpha - m\}X_{n,m+1}, \quad (2.2)$$

where $X_{n,1} \leq \dots \leq X_{n,m}$ denote the order statistics of $\mathcal{X}$, then

$$P(X \leq \tilde{\xi}_\alpha) = \alpha + O(n^{-2}). \quad (2.3)$$

Sufficient regularity conditions are that F have two bounded derivatives in a neighbourhood of ξ_α , and $F'(\xi_\alpha) > 0$. The estimator $\tilde{\xi}_\alpha$ is commonly used in practice, e.g. in the S-PLUS statistical software package, although property (2.3) is not so well known.

We may of course interpolate among three adjacent order statistics, defining

$$\hat{\xi}_\alpha = aX_{n,m-1} + bX_{n,m} + cX_{n,m+1}, \quad (2.4)$$

where the constants a , b and c should be chosen as follows. Select m so that $|m - (n + 1)\alpha| \leq (4/3)^{1/2}$. (As long as $2/(n + 1) \leq \alpha \leq (n - 2)/(n + 1)$, the former inequality is always satisfied by three values of m .) Put $C = (n + 1)\alpha - m$,

$$\begin{aligned} a &= \frac{1}{2}\{2 - C \pm (4 - 3C^2)^{1/2}\}, \\ b &= -\{1 \pm (4 - 3C^2)^{1/2}\}, \\ c &= \frac{1}{2}\{2 + C \pm (4 - 3C^2)^{1/2}\}, \end{aligned}$$

where either the plus or the minus sign is used throughout the definitions of a , b and c . (Thus, for each m satisfying the above inequality, there are just two possible choices of (a, b, c) .) Then, assuming that F is continuous and has three bounded derivatives in a neighbourhood of ξ_α , and that $F'(\xi_\alpha) > 0$, we have

$$P(X \leq \hat{\xi}_\alpha) = \alpha + O(n^{-3}). \quad (2.5)$$

A proof is given in Appendix A.1.

One particular choice of m , in conjunction with a corresponding selection of the triple (a, b, c) , is often preferable. To appreciate why, suppose that the sampling distribution is supported on the positive half-line. Then we wish to select a , b and c , not depending on the data, so that, with probability 1, $\hat{\xi}_\alpha > 0$. There is a unique value of m such that the associated C satisfies $0 \leq C < 1$. For this C , select the second sign throughout in the definitions of a , b and c . Then,

$$\hat{\xi}_\alpha = A(X_{n,m+1} - X_{n,m}) + X_{n,m} + B(X_{n,m} - X_{n,m-1}),$$

where $A = \frac{1}{2}\{2 + C - (4 - 3C^2)^{1/2}\}$ and $B = \frac{1}{2}\{C + (4 - 3C^2)^{1/2} - 2\}$ are both non-negative. Hence, $\hat{\xi}_\alpha$ is a non-negative linear combination of non-negative quantities, implying that $\hat{\xi}_\alpha$ is non-negative. Only for the specified choice of A , B and C does it have this property.

Result (2.5) shows that by interpolation among three order statistics we may reduce the

coverage error to $O(n^{-3})$. However, interpolation among four order statistics is not adequate to give a prediction interval with a coverage error of order n^{-4} ; that requires five order statistics. A proof is given in Appendix A.1, where the case of higher orders of interpolation is discussed. An interval based on interpolation among five order statistics is arguably based too closely on asymptotic properties. Moreover, its coefficients are complex to determine, and the method is not aesthetically attractive. Therefore, interpolation among three order statistics is arguably the practical limit for such procedures.

It may be shown that the variances of these interpolation-based estimators are all a little larger than the variance of $\tilde{\xi}_\alpha$, but only in terms of order n^{-2} . To first order, the variance of each estimator equals $n^{-1}v$, where $v = \alpha(1 - \alpha)/f(\xi_\alpha)$ and $f = F'$ is assumed to be continuous and positive at ξ_α .

2.2. Bootstrap and jackknife methods for calibration

One approach to improving the coverage accuracy is to use bootstrap or jackknife methods to construct an estimator $\hat{\pi}(\beta)$ of the true coverage of intervals such as $(-\infty, \tilde{\xi}_\beta)$ and $(-\infty, \hat{\xi}_\beta)$, for values of β in the neighbourhood of the desired level α . Then, when constructing the final prediction interval, take its nominal level $\hat{\alpha}$ to equal the value of β for which $\hat{\pi}(\beta) = \alpha$.

On reflection it is clear that the conventional bootstrap method must fail to reduce the order of magnitude of the coverage error, if only because of ties that occur in a general bootstrap resample. (Nevertheless, bootstrap calibration of prediction intervals performs well in parametric problems; see Hall *et al.* (1999).) Strikingly, however, the jackknife approach to calibration also fails, performing no better—in terms of the coverage error—than the rudimentary method based on $\tilde{\xi}_\alpha$; see Hall and Rieck (2000).

In contrast, the *smoothed* bootstrap can be used successfully to calibrate prediction intervals and to reduce the order of coverage error. We base the smoothed bootstrap on a kernel estimator of f ,

$$\hat{f}(x) = (nh)^{-1} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right),$$

where K is a symmetric probability density and h denotes the bandwidth. Write $\hat{\eta}_\alpha$ for either $\tilde{\xi}_\alpha$ or $\hat{\xi}_\alpha$. Conditionally on $\mathcal{X}$, let $\mathcal{X}^\dagger = \{X_1^\dagger, \dots, X_n^\dagger\}$ denote a resample drawn by sampling randomly from the distribution with density $\hat{f}$, and let $\hat{\eta}_\alpha^\dagger$ be the version of $\hat{\eta}_\alpha$ that is computed from $\mathcal{X}^\dagger$. Define

$$\begin{aligned}\hat{\pi}(\beta) &= P(X^\dagger \leq \hat{\eta}_\beta^\dagger | \mathcal{X}), \\ \hat{p}(u) &= P(X^\dagger \leq \hat{\eta}_\alpha^\dagger + u | \mathcal{X}),\end{aligned}$$

where $X^\dagger$ is independent of $\mathcal{X}^\dagger$ and has density $\hat{f}$, conditional on $\mathcal{X}$. Let $\beta = \hat{\alpha}$ equal the solution of the equation $\hat{\pi}(\beta) = \alpha$, and $u = \hat{u}_\alpha$ be the solution of $\hat{p}(u) = \alpha$. We claim that if h is chosen appropriately then either of these approaches to calibration reduces the order of magnitude of the coverage error of the prediction interval $(-\infty, \hat{\eta}_\alpha)$, i.e. if $\hat{\zeta}_\alpha$ denotes either $\hat{\eta}_{\hat{\alpha}}$ or $\hat{\eta}_\alpha + \hat{u}_\alpha$, representing level calibration and additive calibration respectively, then

$$P(X \leq \hat{\zeta}_\alpha) = \begin{cases} \alpha + o(n^{-2}) & \text{if } \hat{\eta}_\alpha = \tilde{\xi}_\alpha, \\ \alpha + o(n^{-3}) & \text{if } \hat{\eta}_\alpha = \hat{\xi}_\alpha. \end{cases}$$

Compare equations (2.3) and (2.5).

This result is almost self-evident when $\hat{f}$ is constructed to produce a good estimator (in asymptotic terms, a consistent estimator) of all those derivatives of f that appear explicitly in the first term in an expansion of the coverage error of the uncalibrated interval. When $\hat{\eta}_\alpha = \bar{\xi}_\alpha$, that term involves derivatives up to the second, and when $\hat{\eta}_\alpha = \hat{\xi}_\alpha$ it involves derivatives up to the third. Therefore, in addition to regularity conditions on K and f , it is sufficient that $n^{-(1/5)+\epsilon} \leq h \leq n^{-\epsilon}$ (when $\hat{\eta}_\alpha = \bar{\xi}_\alpha$) or $n^{-(1/7)+\epsilon} \leq h \leq n^{-\epsilon}$ (when $\hat{\eta}_\alpha = \hat{\xi}_\alpha$), for some $\epsilon > 0$ and all sufficiently large n ; these ranges of bandwidth give consistent estimators of the required derivatives of f .

However, smaller bandwidths, in fact so small as to lead to inconsistent estimates of the relevant derivatives, can be used to effect a greater reduction in the order of coverage error. Heuristically, this is because the operation of taking expectations when defining coverage probability is an implicit smoothing step, and so provides smoothing in addition to that supplied through the bandwidth. Therefore, to avoid net oversmoothing, with consequently lower coverage accuracy, we should use a smaller bandwidth in the explicit smoothing step. In fact, the optimal size of bandwidth is $n^{-1/3}$ in the case $\hat{\eta}_\alpha = \bar{\xi}_\alpha$, and $n^{-1/5}$ when $\hat{\eta}_\alpha = \hat{\xi}_\alpha$. It is shown by Hall and Rieck (2000) that

$$\begin{aligned} P(X \leq \bar{\xi}_\alpha + \hat{u}_\alpha) &= \alpha + O(n^{-2-(2/3)}) & \text{if } h \sim cn^{-1/3}, \\ P(X \leq \hat{\xi}_\alpha + \hat{u}_\alpha) &= \alpha + O(n^{-3-(2/5)}) & \text{if } h \sim cn^{-1/5}, \end{aligned}$$

where $c > 0$ is any positive constant. In Section 3, so that there should be no ambiguity we shall use the notation $\bar{u}_\alpha$ for the additive calibration when it is employed in conjunction with $\bar{\xi}_\alpha$, and $\hat{u}_\alpha$ in the case of $\hat{\xi}_\alpha$.

In practice h would generally be chosen by Monte Carlo experiment, as follows. Given a plausible parametric model for the sampling distribution, e.g. normal with estimated mean and variance, choose h through Monte Carlo experiments so that it minimizes the coverage error, and use this bandwidth in the nonparametric procedure. This approach is the analogue, in the present context, of standard ‘comparative’ methods in kernel density estimation; for the latter, see for example Silverman (1986), page 45 and following feature.

Table 1. Optimal bandwidths[†]

n	$\bar{h}_{0.90}$	$\bar{h}_{0.95}$	$\hat{h}_{0.90}$	$\hat{h}_{0.95}$
12	2.62	—	2.78	—
14	2.40	—	2.46	—
16	2.10	2.62	2.20	2.79
18	1.80	2.26	1.91	2.50
20	1.50	1.96	1.63	2.12
22	1.34	1.70	1.45	1.78
24	1.14	1.50	1.30	1.60
26	1.02	1.33	1.14	1.42
30	0.86	1.13	0.96	1.14
35	0.64	0.90	0.73	0.90
40	0.54	0.71	0.50	0.68
45	0.47	0.52	0.40	0.48
50	0.42	0.41	0.28	0.30

[†]Tabulated are bandwidth values that minimize the coverage error in the case of normal $N(0, 1)$ data, various values of sample size n and nominal coverages $\alpha = 0.90$ and $\alpha = 0.95$. Bandwidths $\bar{h}_\alpha$ and $\hat{h}_\alpha$ are appropriate for additive correction of $\bar{\xi}_\alpha$ and $\hat{\xi}_\alpha$ respectively.

Table 1 gives, for a range of sample sizes, the bandwidth that minimizes the absolute coverage error for additive calibration of $\hat{\xi}_\alpha$ and $\hat{\xi}_\alpha$ when $\alpha = 0.9$ or $\alpha = 0.95$ and the true distribution is $N(0, 1)$. Given a more general distribution, we suggest selecting a bandwidth for $N(0, 1)$ data by linearly interpolating on n , and then rescaling the result by multiplying by the sample standard error. Alternatively, a robust estimator of scale could be used.

3. Numerical properties

We present the results from a simulation study comparing performances of four types of prediction interval: I1, $(-\infty, \hat{\xi}_\alpha]$; I2, $(-\infty, \hat{\xi}_\alpha]$; I3, $(-\infty, \hat{\xi}_\alpha + \bar{u}_\alpha]$; I4, $(-\infty, \hat{\xi}_\alpha]$; I5, $(-\infty, \hat{\xi}_\alpha + \hat{u}_\alpha]$. Here, $\bar{u}_\alpha$ and $\hat{u}_\alpha$ denote the additive smoothed bootstrap calibration corrections associated with intervals I2 and I4 respectively. When constructing intervals I4 and I5 we chose m , the index of the central order statistic, such that $C = \{(n + 1)\alpha - m\} \in [0, 1)$, and selected a , b and c in the definition of $\hat{\xi}_\alpha$ at equation (2.4) by $a = -B$, $c = A$ and $b = 1 - a - c$. (See Section 2.1 for definitions of A and B in terms of C .)

We treated four distributions, each corresponding to a different population model P: D1, standard normal; D2, standard exponential; D3, two-parameter Weibull with shape and scale parameters equal to 2 and 1 respectively; D4, standard log-normal. Our results for each population model were based on 5000 simulations. For the additively calibrated intervals I3 and I5 we used the Epanechnikov kernel, given by Silverman (1986), page 143 and following feature. An additional 2000 simulations were used to calculate smoothed bootstrap quantities. To calibrate intervals I3 and I5 we used a bandwidth chosen by interpolation and calibration from Table 1, as discussed in Section 2.2.

For any prediction interval $\mathcal{I}$ the coverage error is defined as the difference between the coverage probability of $\mathcal{I}$ and the nominal coverage: $P(X \in \mathcal{I}) - \alpha$. Table 2 gives approximate coverage errors of the prediction intervals I1–I5 for various sample sizes when the nominal coverage was $\alpha = 0.9$. By comparing these results it can be seen that coverage error is reduced by replacing the two-point interpolation interval I2 by the three-point

Table 2. Coverage errors†

Interval	Results for the following values of n and prediction models D1–D4:							
	$n = 12$				$n = 14$			
	D1	D2	D3	D4	D1	D2	D3	D4
I1	−0.054	−0.051	−0.051	−0.055	−0.035	−0.033	−0.033	−0.033
I2	0.0082	0.011	0.010	0.012	0.0068	0.011	0.010	0.014
I3	−0.0023	0.0063	0.0022	0.0082	−0.0025	0.0054	0.0012	0.0095
I4	0.0044	0.0076	0.0068	0.0081	0.0022	0.0061	0.0053	0.0088
I5	−0.0017	0.0048	0.0018	0.0058	−0.0019	0.0032	0.0034	0.0063
	$n = 16$				$n = 18$			
	D1	D2	D3	D4	D1	D2	D3	D4
I1	−0.018	−0.018	−0.017	−0.017	−0.0049	−0.0036	−0.0023	−0.0046
I2	0.0062	0.0090	0.0080	0.013	0.0032	0.0059	0.0059	0.0073
I3	−0.00078	0.0038	0.0020	0.0083	0.00038	0.0034	0.0021	0.0046
I4	0.0020	0.0039	0.0046	0.0072	0.0012	0.0033	0.0039	0.0037
I5	−0.00042	0.0031	0.0013	0.0050	0.00061	0.0025	0.0012	0.0026

†Values of true coverage minus nominal coverage, the latter being $\alpha = 0.9$, are tabulated for prediction intervals I1–I5 when the population model P is given by D1–D4, and the sample size n is given by 12, 14, 16 and 18.

interval I4. Calibration is also an effective tool for reducing the coverage error when the sample size is small.

An alternative way of depicting coverage accuracy is to plot it against the nominal coverage α for a given sample size. For this investigation the bandwidth for intervals I3 and I5 is fixed for all α and is given simply by $n^{-1/3}$ and $n^{-1/5}$ respectively, where n denotes the sample size. In the case of interval I1 this gives a saw-tooth graph, with coverage error fluctuating erratically in the range from -0.08 to 0 as α increases from 0.2 to 0.9 , for each of distributions D1–D4. The jagged shape of the graph is a consequence of the discontinuous nature of the integer part function, which plays a prominent role in the definition of interval I1. The graph is substantially and progressively smoother for intervals I2–I5. Comparing the graphs for intervals I2 and I4 against those for the calibrated intervals I3 and I5 respectively, it can be seen that the amplitudes of fluctuations are further mitigated in the transition from the former to the latter. Calibration is particularly effective at improving the coverage accuracy when α is close to 0 or 1 .

Appendix A: Technical arguments

A.1. Proof of result (2.5)

Assume that F has three bounded derivatives in a neighbourhood of the α th quantile ξ_α , and that $f = F'$ is bounded away from 0 there. In conducting the proof we shall suppose in addition that F is continuous on the whole real line; this may be assumed without loss of generality, since the probability that one of $X_{m,n-1}$, $X_{m,n}$ and $X_{m,n+1}$ lies outside the neighbourhood where F has three bounded derivatives is exponentially small as a function of n . Initially, suppose of m only that $2 \leq m \leq n-1$, and of a , b and c only that they are bounded in n and satisfy $a + b + c = 1$. Write $U_{n,m} = F(X_{n,m})$, so that $U_{n,1} \leq \dots \leq U_{n,n}$ are the order statistics of a sample from the uniform distribution on $(0, 1)$. If $\hat{\xi}_\alpha$ is given by equation (2.4) then, by Taylor expansion,

$$F(\hat{\xi}_\alpha) = F(X_{n,m}) + \{a(X_{n,m-1} - X_{n,m}) + c(X_{n,m+1} - X_{n,m})\}f(X_{n,m}) \\ + \frac{1}{2}\{a(X_{n,m-1} - X_{n,m}) + c(X_{n,m+1} - X_{n,m})\}^2 f'(X_{n,m}) + O_p(n^{-3}).$$

Writing $X_{n,m+1}$ as $F^{-1}(U_{n,m} + U_{n,m+1} - U_{n,m})$ we see that

$$X_{n,m+1} = X_{n,m} + (U_{n,m+1} - U_{n,m})f(X_{n,m})^{-1} - \frac{1}{2}(U_{n,m+1} - U_{n,m})^2 f'(X_{n,m})f(X_{n,m})^{-3} + O_p(n^{-3}),$$

and an analogous expansion holds for $X_{n,m-1}$. Therefore,

$$F(\hat{\xi}_\alpha) = U_{n,m} + a(U_{n,m-1} - U_{n,m}) + c(U_{n,m+1} - U_{n,m}) + \frac{1}{2}[a(U_{n,m-1} - U_{n,m}) + c(U_{n,m+1} - U_{n,m})]^2 \\ - a(U_{n,m-1} - U_{n,m})^2 - c(U_{n,m+1} - U_{n,m})^2]f'(X_{n,m})f(X_{n,m})^{-2} + R_n, \quad (\text{A.1})$$

where

$$R_n = (|U_{n,m-1} - U_{n,m}|^3 + |U_{n,m+1} - U_{n,m}|^3)S_n$$

and $S_n = O_p(1)$. Since F has three bounded derivatives in a neighbourhood of ξ then it may be proved that for a constant $B > 0$, depending only on F ,

$$P(|S_n| > B) = O(n^{-\lambda}) \quad (\text{A.2})$$

for all $\lambda > 0$.

Let $\mathcal{A}_n$ denote the event that $|S_n| \leq B$. Multiplying both sides of equation (A.1) by the indicator of $\mathcal{A}_n$, taking the expectation, noting that $E(U_{n,i}) = i/(n+1)$ and all polynomial moments of the variables $n(U_{n,i+1} - U_{n,i})$, for $i = -1, 0, 1$, equal the corresponding moments of Z_i plus terms of order n^{-1} , where the random variables Z_i are independent and exponentially distributed, and using equation (A.2) to bound remainder terms where $\mathcal{A}_n$ fails, it may be proved that

$$P(X \leq \hat{\xi}_\alpha) = \frac{m+c-a}{n+1} + \frac{1}{2n^2} \{(a-c)^2 + (a^2 + c^2) - 2(a+c)\} A_n + O(n^{-3}), \quad (\text{A.3})$$

where A_n depends on the sampling distribution but not on a or c . Put $C \equiv (n+1)\alpha - m$. In view of equation (A.3), to obtain $O(n^{-3})$ coverage accuracy we require $c-a = C$ and $(a-c)^2 + a^2 + c^2 - 2(a+c) = 0$. Solving these two equations for a we see that $a = \frac{1}{2}\{2-C \pm (4-3C^2)^{1/2}\}$. Taking $c = a + C$ and $b = -1 \mp (4-3C^2)^{1/2}$ we see that the desired result (2.5) follows from equation (A.3).

A.2. Four-point interpolation fails to give $O(n^{-4})$ accuracy

Assume that F is continuous on the real line, has four bounded derivatives in a neighbourhood of the α th quantile ξ_α and that $f = F'$ is bounded away from 0 there. For notational simplicity we outline the proof in the case of interpolation among four adjacent order statistics; the non-adjacent case is similar. We may write the interpolant as $\hat{\xi}_\alpha = aX_{n,m} + bX_{n,m+1} + cX_{n,m+2} + dX_{n,m+3}$, where $m = (n+1)\alpha + O(1)$ and m and the bounded constants a, b, c and d should be chosen, depending on n but not on F , in such a way that the coverage error equals $O(n^{-4})$. It is clear that $a+b+c+d = 1 + O(n^{-4})$. Without loss of generality the $O(n^{-4})$ term here vanishes. Then, $\hat{\xi}_\alpha = X_{n,m} + \Delta$ where

$$\Delta = b(X_{n,m+1} - X_{n,m}) + c(X_{n,m+2} - X_{n,m}) + d(X_{n,m+3} - X_{n,m}),$$

and

$$F(\hat{\xi}_\alpha) = F(X_{n,m}) + \Delta f(X_{n,m}) + \frac{1}{2}\Delta^2 f'(X_{n,m}) + \frac{1}{6}\Delta^3 f''(X_{n,m}) + O_p(n^{-4}).$$

Writing $G = F^{-1}$ and $\delta_k = U_{n,m+k} - U_{n,m}$ we have

$$X_{n,m+k} - X_{n,m} = \sum_{r=1}^3 \frac{1}{r!} \delta_k^r G^{(r)}(X_{n,m}) + O_p(n^{-4}),$$

and so

$$\begin{aligned} F(\hat{\xi}_\alpha) &= F(X_{n,m}) + b\delta_1 + c\delta_2 + d\delta_3 + \frac{1}{2}\{(b\delta_1 + c\delta_2 + d\delta_3)^2 - (b\delta_1^2 + c\delta_2^2 + d\delta_3^2)\} \frac{f'(X_{n,m})}{f(X_{n,m})^2} \\ &\quad + \frac{1}{6}\left[(b\delta_1 + c\delta_2 + d\delta_3)^3 \frac{f''(X_{n,m})}{f(X_{n,m})^3} - 3(b\delta_1 + c\delta_2 + d\delta_3)(b\delta_1^2 + c\delta_2^2 + d\delta_3^2) \frac{f'(X_{n,m})}{f(X_{n,m})^4}\right. \\ &\quad \left. + (b\delta_1^3 + c\delta_2^3 + d\delta_3^3) \left\{ \frac{3f'(X_{n,m})^2}{f(X_{n,m})^4} - \frac{f''(X_{n,m})}{f(X_{n,m})^3} \right\}\right] + O_p(n^{-4}). \end{aligned} \quad (\text{A.4})$$

Note that

$$E\{F(X_{n,m}) + b\delta_1 + c\delta_2 + d\delta_3\} = \frac{m+b+2c+3d}{n+1}.$$

Moreover, conditional on $X_{n,m}$, the variables $\delta_k/\{1 - F(X_{n,m})\}$ for $k = 1, 2, 3$ have the joint distribution of the k smallest of $n - m$ random variables that are uniformly distributed on the interval $(0, 1)$. Denote such a random variable by V_k ; it may be taken to be independent of $X_{n,m}$. In this notation we deduce from equation (A.4), using arguments in Appendix A.1, that

$$\begin{aligned} E\{F(\hat{\xi}_\alpha)\} &= \frac{m+b+2c+3d}{n+1} + \frac{1}{2} \gamma_1(n) E\{(bV_1 + cV_2 + dV_3)^2 - (bV_1^2 + cV_2^2 + dV_3^2)\} \\ &\quad + \frac{1}{6} [\gamma_3(n) E\{(bV_1 + cV_2 + dV_3)^3\} - 3 \gamma_2(n) E\{(bV_1 + cV_2 + dV_3)(bV_1^2 + cV_2^2 + dV_3^2)\} \\ &\quad + \{3 \gamma_2(n) - \gamma_3(n)\} E(bV_1^3 + cV_2^3 + dV_3^3)] + O(n^{-4}), \end{aligned} \quad (\text{A.5})$$

where

$$\left. \begin{aligned} \gamma_1(n) &= E \left[\frac{f'(X_{n,m})}{f(X_{n,m})^2} \{1 - F(X_{n,m})\}^2 \right], \\ \gamma_2(n) &= E \left[\frac{f'(X_{n,m})^2}{f(X_{n,m})^4} \{1 - F(X_{n,m})\}^3 \right], \\ \gamma_3(n) &= E \left[\frac{f''(X_{n,m})}{f(X_{n,m})^3} \{1 - F(X_{n,m})\}^3 \right]. \end{aligned} \right\} \quad (\text{A.6})$$

The terms with coefficients $\frac{1}{2}$ and $\frac{1}{6}$ on the right-hand side of equation (A.5) are of orders n^{-2} and n^{-3} respectively.

Since the variable m changes only in integer increments then it is not independent in an algebraic sense. For the coverage error to equal $O(n^{-4})$, for general distributions and general values of α , the first term on the right-hand side of equation (A.5), which does not depend on the sampling distribution, must equal $\alpha + O(n^{-4})$. Without loss of generality it equals α exactly. This gives one equation that the variables b , c and d must satisfy, as functions of α and m . The expectations involving V_1 , V_2 and V_3 do not depend on the sampling distribution, but the quantities $\gamma_j(n)$ do. Hence, setting terms of orders n^{-2} and n^{-3} equal to at most $O(n^{-4})$ requires three other equations in b , c and d . It is shown in Rieck (2001) that the resulting four equations are, up to terms of order n^{-4} , independent in the three variables b , c and d , and so for general α there do not exist m and a , b , c and d such that $E\{F(\hat{\xi}_\alpha)\} = \alpha + O(n^{-4})$.

If the interpolation is extended to five, rather than three, order statistics then an expansion of coverage error up to a remainder of order n^{-4} can be expressed in terms of five equations in five unknowns; this time the equations admit a simultaneous solution. Reducing the coverage error further, to $O(n^{-5})$, requires another three equations, for the unknowns $f'''(\xi_\alpha)/f(\xi_\alpha)^4$, $f'(\xi_\alpha)^3/f(\xi_\alpha)^6$ and $f'(\xi_\alpha)f''(\xi_\alpha)/f(\xi_\alpha)^5$. Therefore, interpolation among eight order statistics is required on this occasion.

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